

Matrices in Geometry - 2.10.46

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Problem Statement

Given vectors $\mathbf{V} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$, $\mathbf{W} = \mathbf{i} + 3\mathbf{k}$ and $\mathbf{U}$ is a unit vector, find the maximum value of the scalar triple product $[\mathbf{U} \ \mathbf{V} \ \mathbf{W}]$.

Vector Representations

Let us express the vectors in matrix form:

$$\mathbf{v} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} U_x \\ U_y \\ U_z \end{pmatrix}, \quad \|\mathbf{u}\| = 1. \quad (1)$$

Scalar Triple Product as a Determinant

The scalar triple product S is the determinant of the matrix formed by $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{W}$ as columns:

$$S = \det \begin{pmatrix} U_x & 2 & 1 \\ U_y & 1 & 0 \\ U_z & -1 & 3 \end{pmatrix}. \quad (2)$$

Expanding along the first column:

$$S = U_x \det \begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix} - U_y \det \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix} + U_z \det \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}. \quad (3)$$

Determinant Evaluation

Calculate the minors:

$$\det \begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix} = (1)(3) - (0)(-1) = 3, \quad (4)$$

$$\det \begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix} = (2)(3) - (1)(-1) = 6 + 1 = 7, \quad (5)$$

$$\det \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} = (2)(0) - (1)(1) = -1. \quad (6)$$

Substitute back:

$$S = 3U_x - 7U_y - U_z. \quad (7)$$

Maximizing the Scalar Triple Product

Since $\mathbf{U} = \begin{pmatrix} U_x \\ U_y \\ U_z \end{pmatrix}$ is a unit vector,

$$U_x^2 + U_y^2 + U_z^2 = 1. \quad (8)$$

Define

$$\mathbf{A} = \begin{pmatrix} 3 \\ -7 \\ -1 \end{pmatrix}. \quad (9)$$

The scalar triple product reduces to the dot product

$$S = \mathbf{U}^\top \mathbf{A}. \quad (10)$$

The maximum value of $\mathbf{U}^\top \mathbf{A}$ subject to $\|\mathbf{U}\| = 1$ is

$$\max S = \|\mathbf{A}\|. \quad (11)$$

Norm Calculation

Compute the norm $\|\mathbf{A}\|$:

$$\|\mathbf{A}\| = \sqrt{3^2 + (-7)^2 + (-1)^2} = \sqrt{9 + 49 + 1} = \sqrt{59}. \quad (12)$$

Final Result

Hence, the **maximum value** of the scalar triple product is

$$\max [\mathbf{U} \quad \mathbf{V} \quad \mathbf{W}] = \sqrt{59}. \quad (13)$$

C code

```
// Define vectors V and W
double V[3] = {2, 1, -1};
double W[3] = {1, 0, 3};

// Array to store cross product
double cross_V_W[3];

// Function to compute cross product
void compute_cross_product() {
    cross_V_W[0] = V[1]*W[2] - V[2]*W[1];
    cross_V_W[1] = V[2]*W[0] - V[0]*W[2];
    cross_V_W[2] = V[0]*W[1] - V[1]*W[0];
}
```

C code

```
// Functions to access cross product components
double* get_cross_product() {
    return cross_V_W;
}
```

Python code

```
import numpy as np
import matplotlib.pyplot as plt

# Define vectors V and W
V = np.array([2, 1, -1])
W = np.array([1, 0, 3])

# Compute cross product
cross_V_W = np.cross(V, W)

# Plotting the vectors and their cross product
origin = np.zeros(3)

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
```

Python code

```
# Plot V
```

```
ax.quiver(*origin , *V, color='blue ' , label='V')
```

```
# Plot W
```

```
ax.quiver(*origin , *W, color='green ' , label='W')
```

```
# Plot cross product of V and W
```

```
ax.quiver(*origin , *cross_V_W , color='red ' , label='V x W')
```

```
# Setting plot limits and labels
```

```
ax.set_xlim([-3, 3])
```

```
ax.set_ylim([-3, 3])
```

```
ax.set_zlim([-3, 3])
```

Python code

```
ax.set_xlabel('X-axis')
ax.set_ylabel('Y-axis')
ax.set_zlabel('Z-axis')
ax.set_title('Vectors V, W and Cross Product  $V \times W$ ')

ax.legend()
plt.show()
```

Plot

Vectors V , W and Cross Product $V \times W$

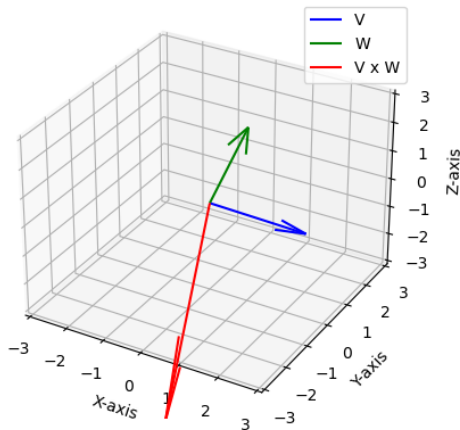


Figure: 3D Plot